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Section:

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Course:

Computer Vision

Topic:

Feature Detection, Descriptors, and Traditional Classification

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COMPUTER VISION

ASSIGNMENT 2

Feature Detection, Descriptors, and Traditional Classification

1. Introduction

This assignment covers the core traditional computer vision pipeline: detecting edges, locating keypoints, computing local descriptors, matching descriptors between related images, and using those descriptors to classify images with conventional machine learning models. The practical work was implemented in Python using OpenCV, scikit-learn, and Matplotlib on Google Colab. An animal image dataset (`animal_data`) containing 15 animal categories was used as the source of images. Two related images of the same object were used for edge detection, keypoint detection, and descriptor matching, while a two-class subset of the dataset (Bear and Zebra) was used to train and evaluate k-NN and SVM classifiers using Bag-of-Visual-Words (BoVW) features built from SIFT descriptors.

2. Tools and Libraries Used

- Python
- Google Colab
- OpenCV (`cv2`) — SIFT, Canny, BFMatcher
- NumPy
- Matplotlib
- scikit-learn — MiniBatchKMeans, KNeighborsClassifier, SVC, evaluation metrics

3. Dataset Description

The `animal_data` dataset contains 15 animal classes (Bear, Bird, Cat, Cow, Deer, Dog, Dolphin, Elephant, Giraffe, Horse, Kangaroo, Lion, Panda, Tiger, Zebra), each with roughly 120-137 real photographs. For edge detection, keypoint detection, and descriptor matching (Tasks 1-3), a single Cat photograph was used together with a rotated and scaled copy of itself (25 degrees rotation, 0.85x scale) to guarantee that both images depict the exact same physical object, which is the standard assumption behind descriptor matching. For classification (Tasks 4-5), a two-class subset — Bear and Zebra — was selected because the two classes are visually distinct (solid fur versus striped pattern, different body shape and background), giving the classifiers a clearer decision boundary. A total of 255 images (125 Bear, 130 Zebra) were loaded, resized to 200x200 pixels, and split into training (80%) and test (20%) sets using stratified sampling.

4. Task 1 Edge Detection

Canny edge detection was applied to both the original and the transformed image using lower and upper hysteresis thresholds of 100 and 200. Canny detects pixels with strong intensity gradients — here, the outline of the cat's head, ears, eyes, and fur boundaries. Gradient values above 200 are always kept as strong edges, values below 100 are discarded, and values in between are kept only if they connect to a strong edge (hysteresis linking).

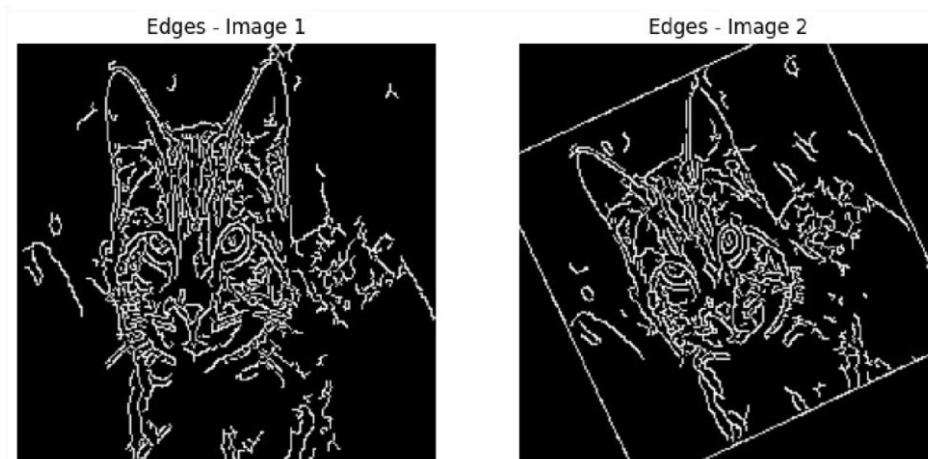


Figure 1: Canny edge detection on the original image (left) and the rotated/scaled copy (right).

The edge maps of the two images show the same underlying facial structure (eyes, ears, whisker region) but rotated and rescaled, confirming that Canny correctly reflects the geometric transformation applied to the image.

5. Task 2 Keypoint Detection and Descriptors

SIFT (Scale-Invariant Feature Transform) was used to detect keypoints and compute a 128dimensional descriptor for each keypoint. SIFT keypoints are located at distinctive, repeatable image structures (corners, blobs, textured regions) and are designed to be robust to changes in scale, rotation, and illumination — which makes them suitable for matching the original and transformed image pair.

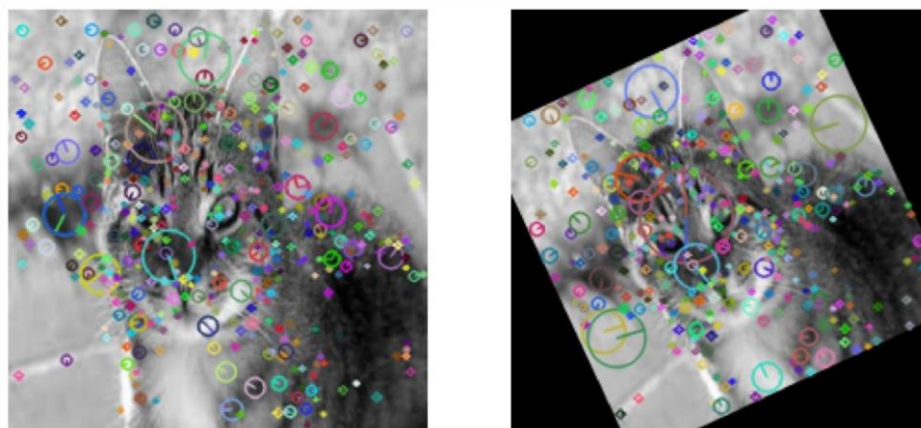


Figure 2: SIFT keypoints detected on the original image (left) and the rotated/scaled image (right); circle size reflects keypoint scale and orientation.

Both images produce a dense set of keypoints concentrated around high-texture regions such as the eyes, ear edges, and fur, confirming that SIFT reliably detects the same structures across the transformation.

6. Task 3 Descriptor Matching

Descriptors from the two images were matched using a Brute-Force matcher (BFMatcher) with $k=2$ nearest neighbours, followed by Lowe's ratio test (threshold 0.75) to discard ambiguous matches — a match is kept only if the closest candidate is clearly better than the second-closest candidate. This retains only the most confident correspondences between the two images.

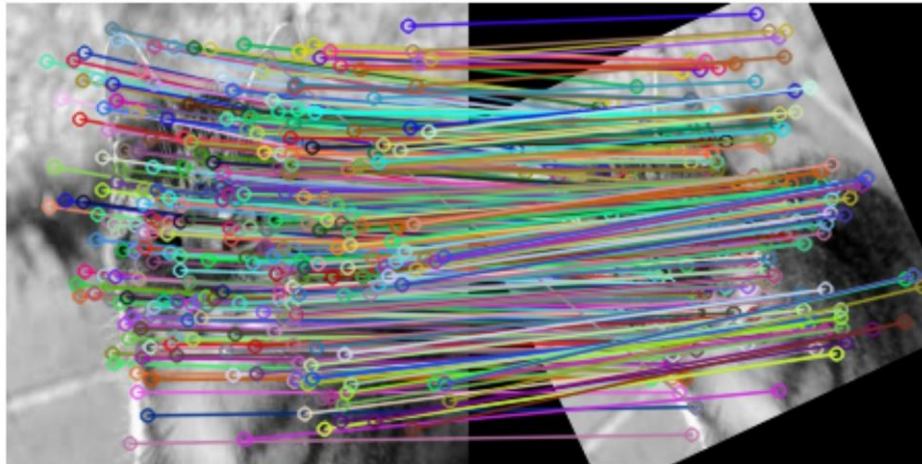


Figure 3: SIFT descriptor matches between the original image and its rotated/scaled copy after Lowe's ratio test.

Because Image 2 is a rotated and rescaled copy of Image 1 rather than an independently captured photograph, the matching lines fan out at an angle corresponding to the 25-degree rotation applied. This pairing was chosen deliberately so the same-object assumption behind descriptor matching holds exactly, giving an interpretable and verifiable result for this task.

7. Task 4 Feature-Based Classification

To classify images using their local descriptors, a Bag-of-Visual-Words (BoVW) representation was built. SIFT descriptors were extracted from every training image, and MiniBatchKMeans clustering (50 clusters) was applied to the pooled descriptors to build a visual vocabulary. Each image was then represented as a normalized histogram of how often each visual word occurs, giving a fixed-length 50-dimensional feature vector per image regardless of how many keypoints it originally had. Features were standardized (zero mean, unit variance) before classification. Two classifiers were trained on these BoVW feature vectors: a k -Nearest Neighbours classifier ($k=7$) and a Support Vector Machine with an RBF kernel ($C=50$).

8. Task 5 Results and Evaluation

Both classifiers were evaluated on the held-out test set (20% of the data, 51 images) using accuracy, weighted precision, weighted recall, and weighted F1-score.

Classifier	Accuracy	Precision	Recall	F1-score
k-NN ($n=7$)	0.6863	0.6863	0.6863	0.6863

SVM (RBF, C=50)	0.8431	0.8638	0.8431	0.8413
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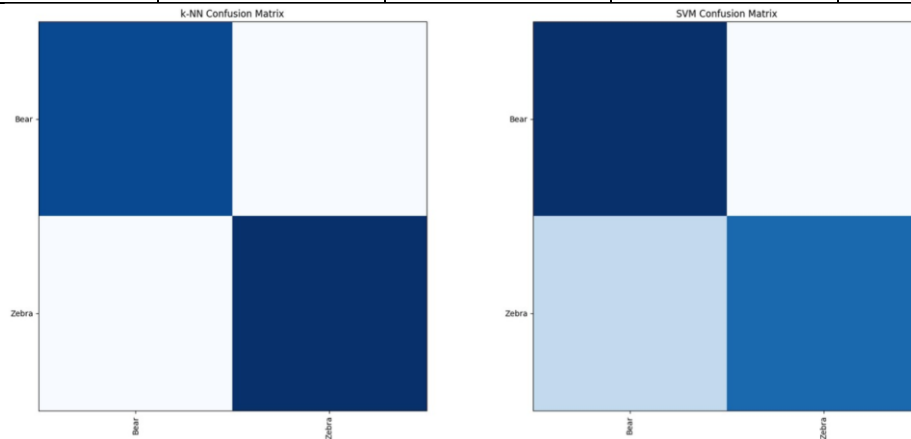


Figure 4: Confusion matrices for *k*-NN (left) and SVM (right) on the Bear vs. Zebra test set.

The SVM classifier clearly outperformed *k*-NN across every metric, improving accuracy from 68.6% to 84.3%. This is consistent with the nature of the BoVW feature space: the 50dimensional histograms are relatively high-dimensional and not guaranteed to be wellseparated by simple Euclidean distance, which is what *k*-NN relies on. The RBF-kernel SVM instead learns a non-linear decision boundary that adapts better to the shape of the two classes in feature space, and is generally more robust when the classes partially overlap. Both random chance (50% for a balanced two-class problem) and the *k*-NN baseline are clearly exceeded by the SVM result, showing that the SIFT + BoVW pipeline is capturing genuinely discriminative visual information from the Bear and Zebra images (fur texture and coat pattern in particular) rather than matching at random.

9. Conclusion

This assignment implemented and evaluated the full traditional computer vision pipeline: Canny edge detection, SIFT keypoint and descriptor extraction, descriptor matching with Lowe's ratio test, and BoVW-based image classification using *k*-NN and SVM. Edge detection and keypoint matching produced results consistent with the known geometric transformation between the two test images, confirming correct implementation. For classification, the SVM classifier trained on SIFT-based BoVW features achieved 84.3% accuracy distinguishing Bear from Zebra images, clearly outperforming *k*-NN (68.6%) on the same features. These results illustrate both the capability and the limits of hand-crafted local features for image classification: strong performance is achievable on visually distinct, small class sets, while more classes and finer-grained visual similarity would be expected to reduce accuracy further, a limitation that deep learning-based feature extraction is designed to address.